$$E(X) = \sum x_i p_i$$

$$= 1\times\frac{1}{6}+2\times\frac{1}{6}+.....+6\times\frac{1}{6}$$

$$= \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = \frac{7}{2}$$

Similarly,

$$E(Y) = \frac{7}{2}$$

Thus the expectation of the numbers obtained on two dices.

$X+Y$ takes the values 2, 3…12 with their probability given by

x	2	3	4	5	6	7	8	9	10	11	12
$P(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

$$E(X+Y) = 2\times\frac{1}{36}+3\times\frac{2}{36}+...+12\times\frac{1}{36}$$

$$= \frac{2+6+12+20+30+42+40+36+30+22+12}{36} = 7$$

$$E(X)+E(Y) = \frac{7}{2}+\frac{7}{2} = 7$$

$$\therefore \quad E(X+Y) = E(X) + E(Y)$$

Example 9.16

Let X and Y are two random variables with p.d.f given by

$$f(x,y) = \begin{cases} 4xy, & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$$

and their marginal density functions as

$$f(x) = \begin{cases} 2x, & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$$

and $g(y) = \begin{cases} 2y, & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$

prove that $E(X+Y) = E(X)+E(Y)$

Solution:

$$E(X+Y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x+y)f(x,y)\,dxdy$$

$$= \int_0^1\int_0^1 (x+y)4xy\,dxdy$$